An Introduction to Hydraulic Actuators and Circuits

Basics of Hydraulics and Hydraulic Actuator Systems

Outline

- Hydraulics Basics
- Hydraulics Actuator System

Learning Objectives

- What will you know/be able to do by the end?
  - Explain basic hydraulic principles
  - Identify major hydraulic system parts

Hydraulics Actuation Application



Aerial Work Platforms



Hydraulic Shop Press



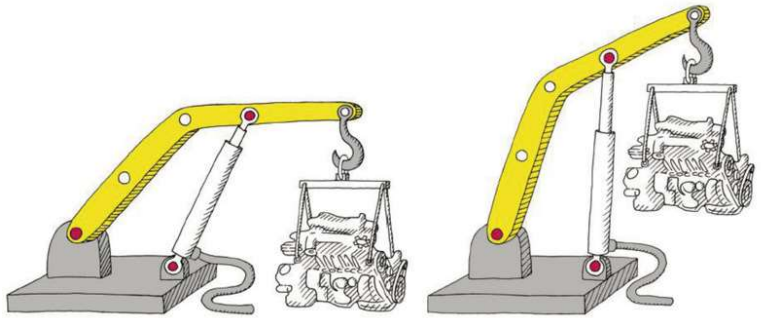
Hydraulic Lift Table Cart

Hydraulics Applications

- Extremely high force output for compact size
- Smooth, precise, reversible motion







Hydraulics

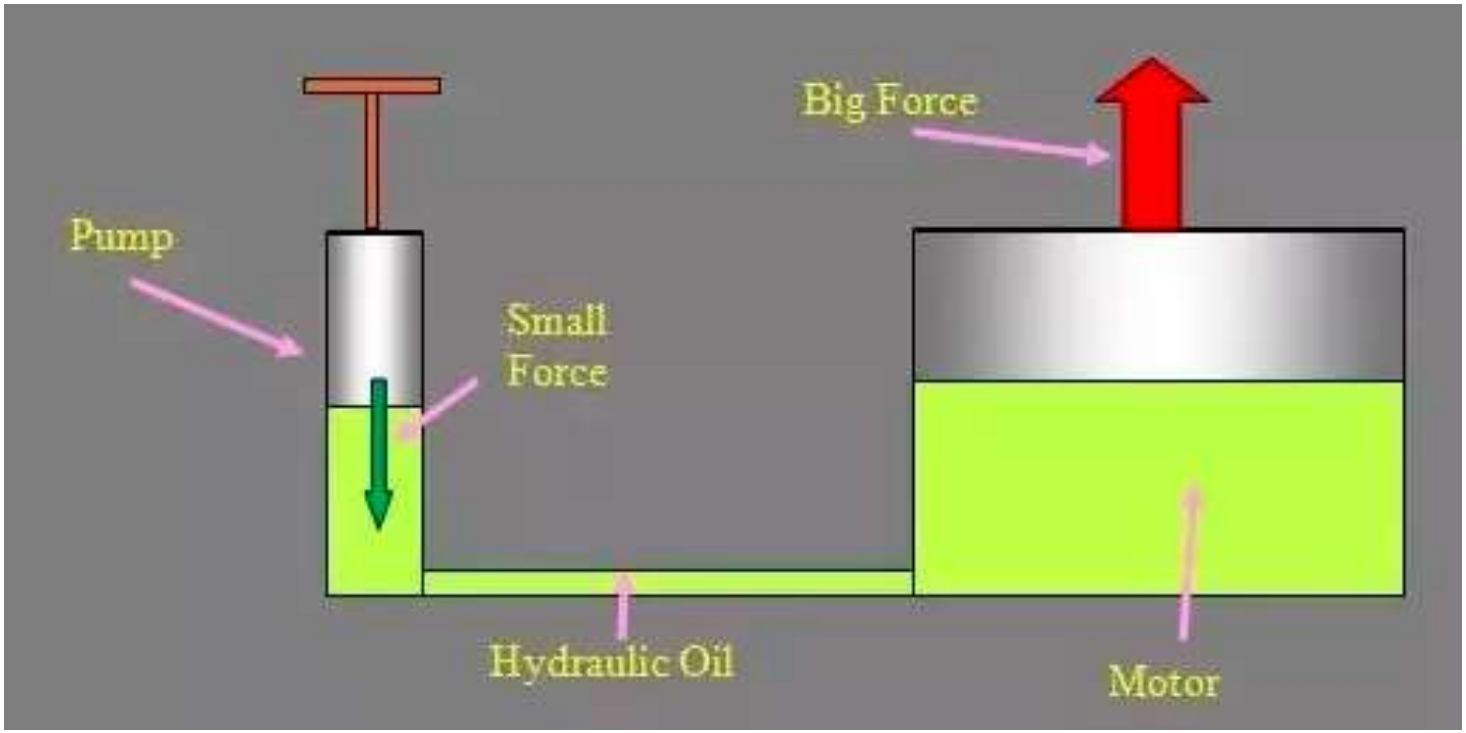
- The word "hydraulics" originates from the Ancient Greek word hydraulikos, which means "of a water organ".

  Hydor: The Greek word for "water".
  Aulos: The Greek word for a "pipe" or "flute".

  Hydraulics = Controlled flow of pressurized fluid

Hydraulics is the branch of fluid power that uses pressurized liquids (typically oil) generate and transmit power for heavy duty applications like applications like cranes and press machinery

Hydraulic System



In **hydraulics-based systems,** mechanical movement is produced by **confined, pumped liquid**, typically through **cylinders moving pistons.**

Basic Principles of Hydraulics

Fluid mechanics provides the theoretical foundation for hydraulics, which focuses on applied engineering using the properties of fluids.

Review:

- Force
- Pressure
- Pascal's Law
- Flow, Flow rate

Force and Pressure

Force (F): A push or pull exerted on an object that can cause it to move, stop, or change direction.

$$F = m \times a$$

where m = mass (kg), a = acceleration (m/s^2)



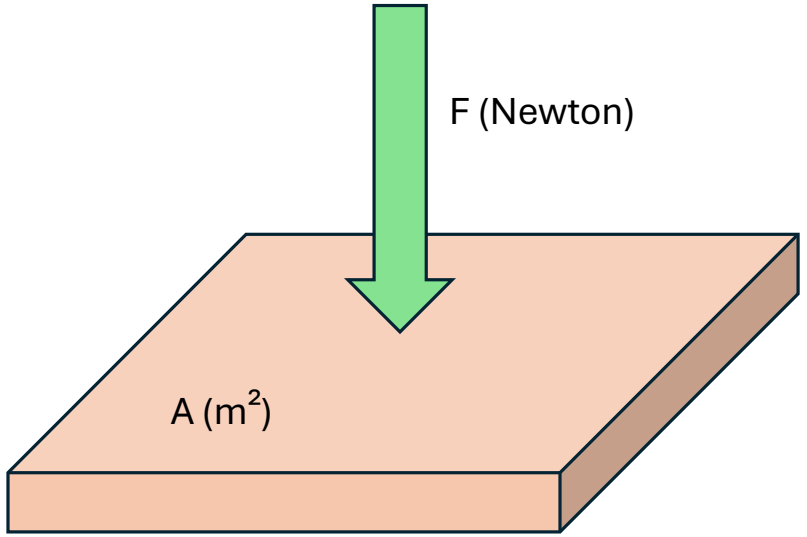
Units:
SI Unit: Newton (N)
1 N=1 $kg.m/s^2$

Pressure

Pressure (P): Pressure is the amount of force (F) applied per unit area (A) of a surface.

$$P = F/A$$

where F = Force (N) and A = Area of cross-section in m^2



Units:

- **Pascal (Pa),** the Pound per Square Inch **(psi)**, and the **Bar**
- Pascal (Pa) = N/m^2
- 1 bar = 10^5 Pa = 100 kPa = 14.5 psi

Examples - Find Pressure

Q. *A **force of 100N** is applied over an **area of 5m²** . **<u>What is the pressure?</u>***

P = F / A
P = 100/ 5
P = 20 Pa

Q. A pressure of 55KPa is felt over an area of 8mm². What is the applied Force?

$$F = P \times A$$

$$F = 55,000 \text{ Pa} \times 8 \times 10^{-6} \text{ m}^2$$

$$F = 0.44 \text{ N}$$

$$1 \text{ mm}^2 = 1 \times 10^{-6} \text{ m}^2$$

$$A = 8 \times 10^{-6} \text{ m}^2$$

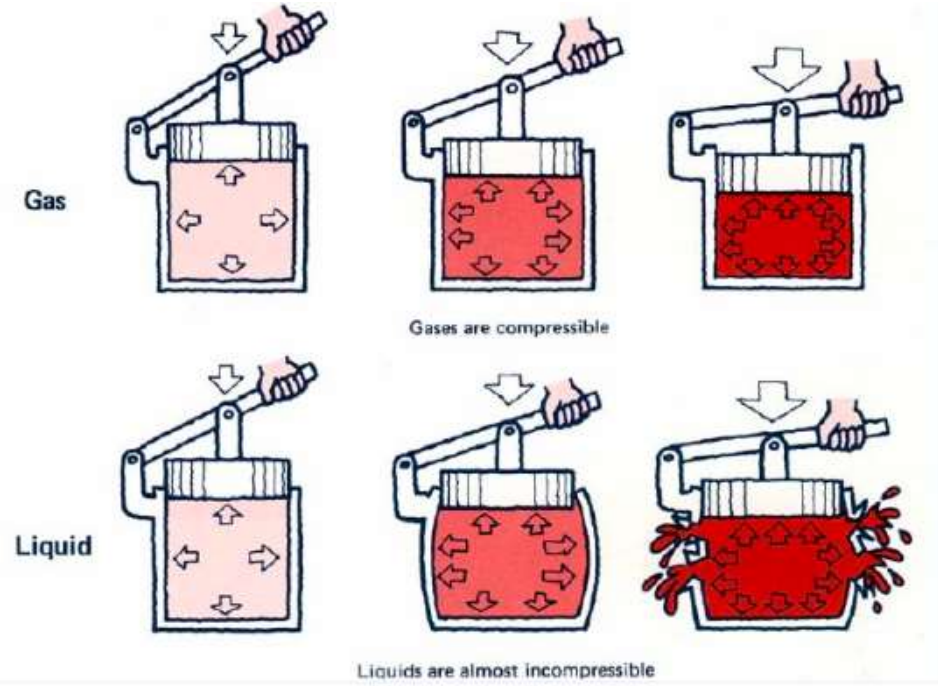
Pressure in Confined Fluids

Compressibility:

- Liquids are nearly incompressible (≈ 0.5 % volume change per 100 bar)
- In gases (air), pressure also spreads but gases are compressible → some energy stored as compression.

Force Transmission:

- **Hydraulic systems** (liquids): instant, precise force transmission.
- **Pneumatic systems** (air): elastic, less precise force transmission.



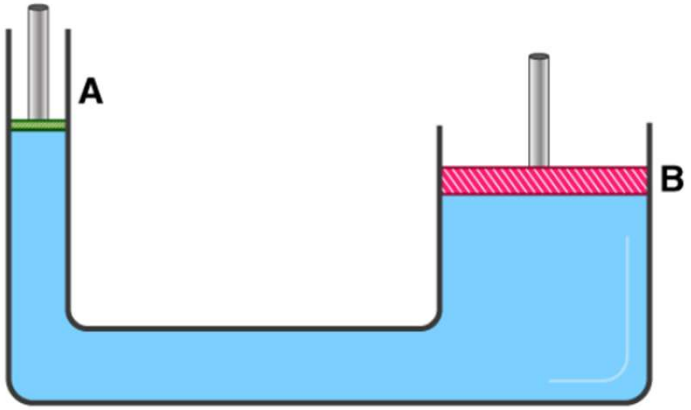
- Equal pressure on all surfaces
- Force acts perpendicular to boundaries

Pascal's Law

Pascal's Law: Pressure applied to a confined fluid is transmitted **equally in all directions through out the fluid.**

So, a force applied at one point on a liquid will spread equally to other points in the liquid.

By confining fluid, we turn a small push into a much larger force, transmitted smoothly and evenly.



Pressure at A = Pressure at B

Hydraulic engineering is founded on Pascal's principle, utilizing the fact that a small force applied over a small cross-sectional area can generate a much larger force over a larger area through pressure transmission.

Pascal Law's Application

A1 = 0. 1 m^2

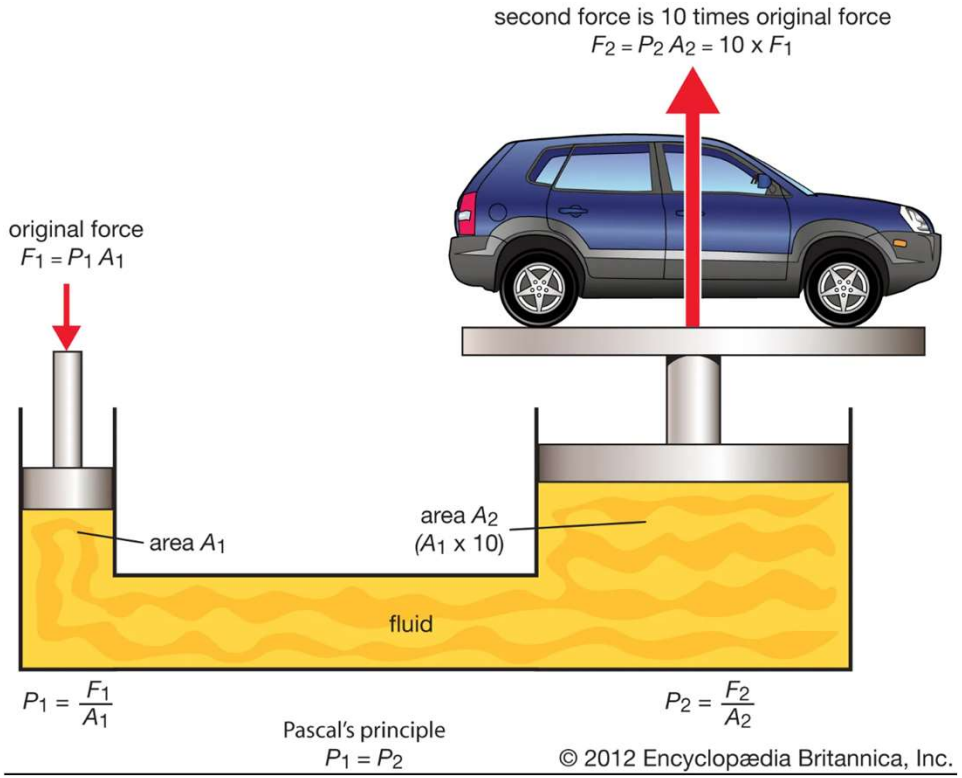
A2 = 1 m^2

Assume we apply **F1 = 20 N**, substituting the value,

P = 200 Pa =2000 N/m^2

Pascal Law: P1 = P 2

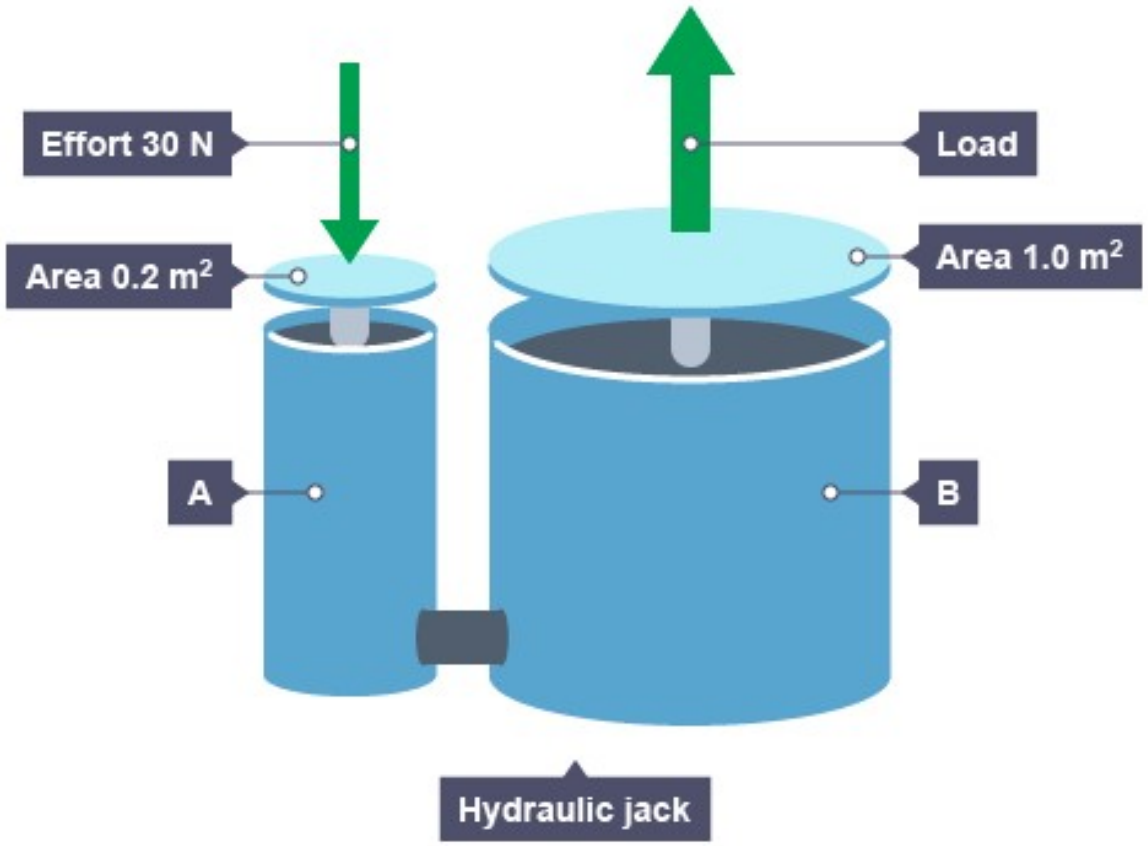
F2 = 200 N



Implication:

- Small input force → large output force (if A2>A1)
- Output moves shorter distance → energy conserved
- ***Hydraulic systems*** *therefore* ***allow smaller forces*** *to be* ***multiplied into bigger forces.***

Example



Step 1: First calculate the pressure of the liquid inside piston A

Step 2: Find the force in Piston B

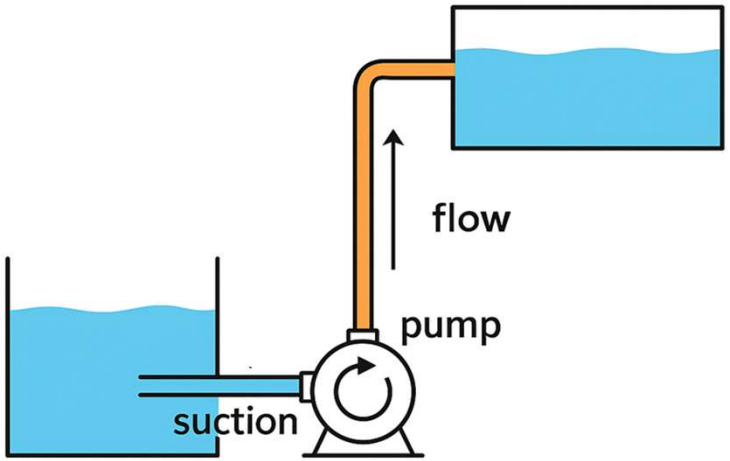
F1/A1 = F2/A2

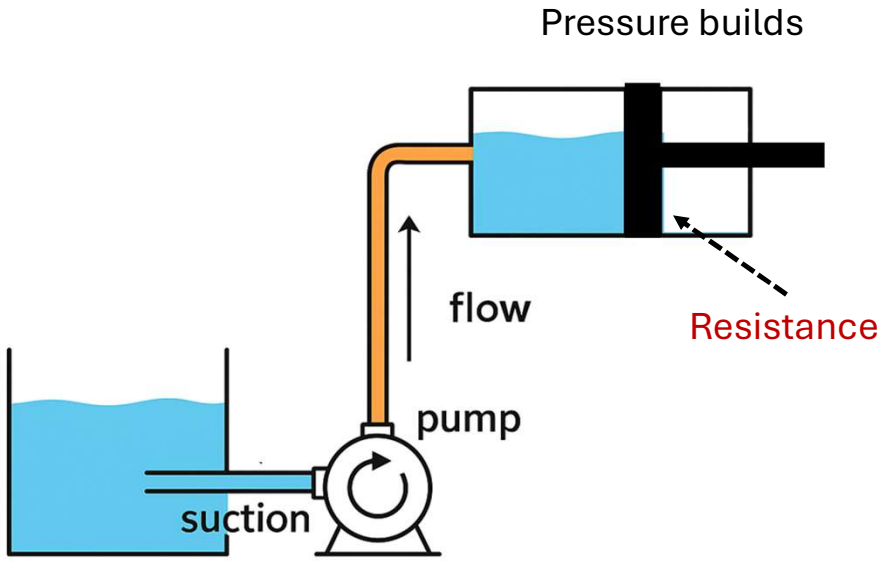
F2 = 150 N

Principles of Flow, Velocity, and Flow Rate

Flow: The *movement of fluid* that gives the actuator its motion.

- Created by the **pump** in the hydraulic system.
- Flow produces **motion**; resistance creates **pressure**.

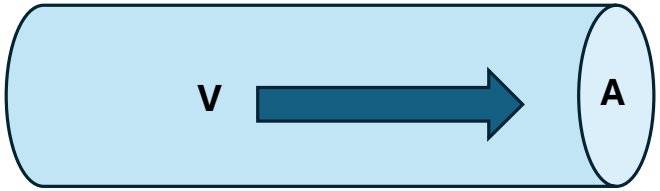




Velocity, and Flow Rate

Velocity (v): Average speed of fluid particles

$$v = \frac{\text{distance}}{\text{time}} \quad [\text{m/s}]$$



where A = pipe cross-sectional area, v = velocity

Velocity tells how fast the fluid is moving inside the lines

Flow Rate (Q): Volume of fluid passing through a pipe per unit time

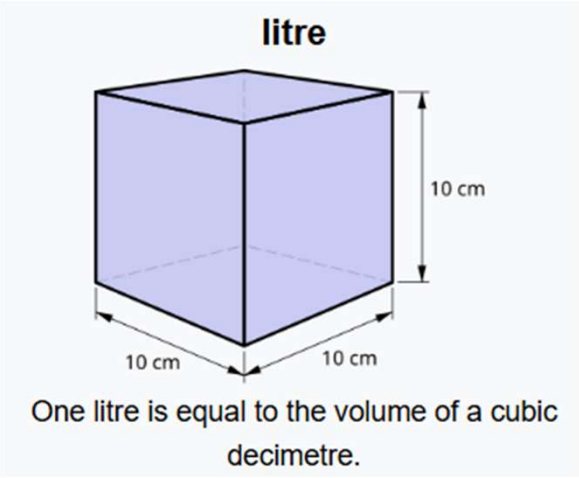
$$Q = A \times v \quad [\text{m}^3/\text{s or L/s}]$$

Flow rate tells us how much fluid passes per second.

Together V and Q determine the **speed of the actuator**

Example Problem: Flow Rate and Velocity

Q. Oil flows through a 30 mm diameter pipe at 60 L/min. Find velocity (v).



$$1\ \text{L} \equiv 1000\ \text{cm}^3 = 0.001\ \text{m}^3$$

$$1\ \text{L/min} = \frac{1}{1000}\ \text{m}^3 \div 60\ \text{s} = 0.00001667$$

$$Q(\text{m}^3/\text{s}) = Q(\text{L/min}) \times 0.00001667$$

Solution

1. Convert flow rate:

$$Q = 60\ \text{L/min} = 60 \times 0.0000167 = 0.0010\ \text{m}^3/\text{s}$$

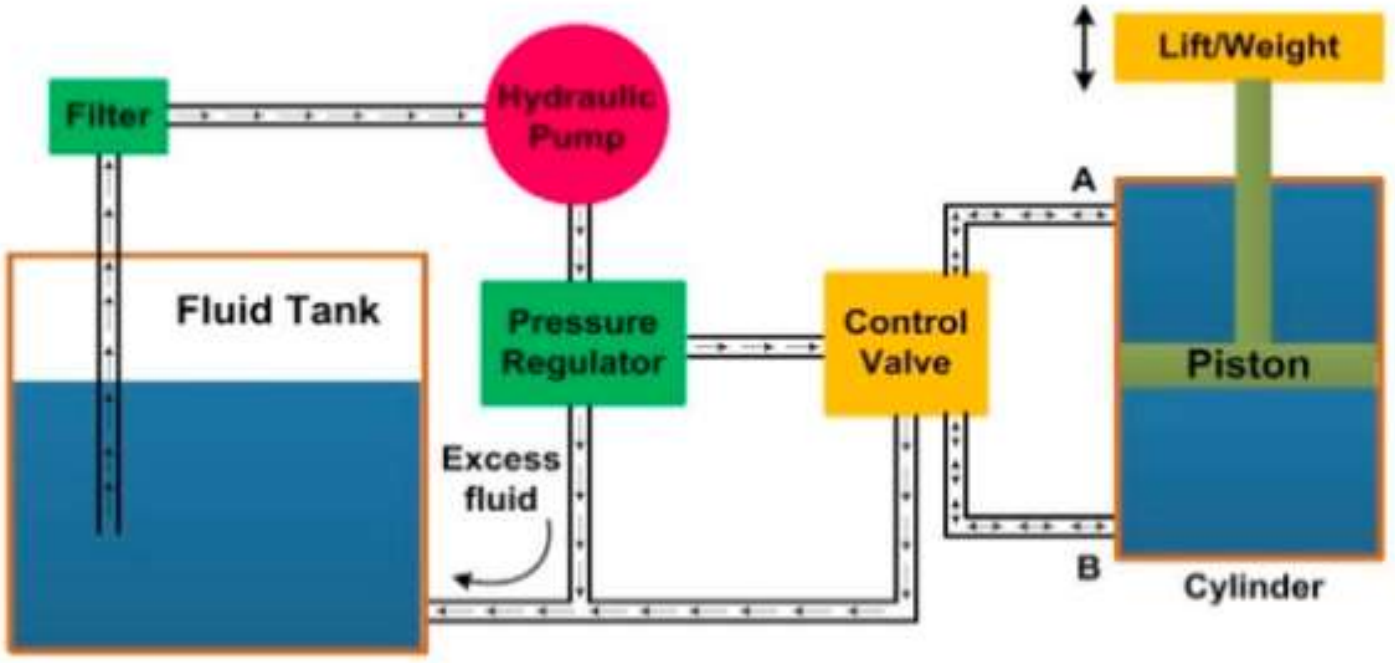
2. Find pipe area:

$$A = \frac{\pi D^2}{4} = \frac{3.14(0.03)^2}{4} = 0.000707\ \text{m}^2$$

3. Calculate velocity:

$$v = \frac{Q}{A} = \frac{0.0010}{0.000707} = 1.41\ \text{m/s}$$

Hydraulic System Overview



Main elements:

1. Reservoir (stores fluid)
2. Pump (creates flow)
3. Valves (control direction, pressure, flow)
4. Actuator (Does work)
5. Return line and filters

Major Components of a Hydraulic Actuator System

- **Hydraulic Pump** – Converts mechanical to fluid power.
- **Hydraulic Valves** – Control flow, pressure, and direction.
- **Hydraulic Actuator** – Convert hydraulic power to linear (cylinder) or rotary motion (motor) .

Reservoir and Fluid



Reservoir (or Tank)

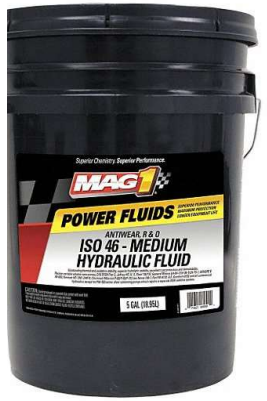
- Stores hydraulic fluid and keeps it clean.
- Provides space for fluid expansion and deaeration.
- Ensures fluid supply to the pump at low pressure.

Hydraulic Fluid

Common fluids: **mineral oil** or **water-based fluids**

- Transmits power (primary function).
- Lubricates moving components.
- Dissipates heat generated in operation.
- Protects components from corrosion and wear





Hydraulic Pump

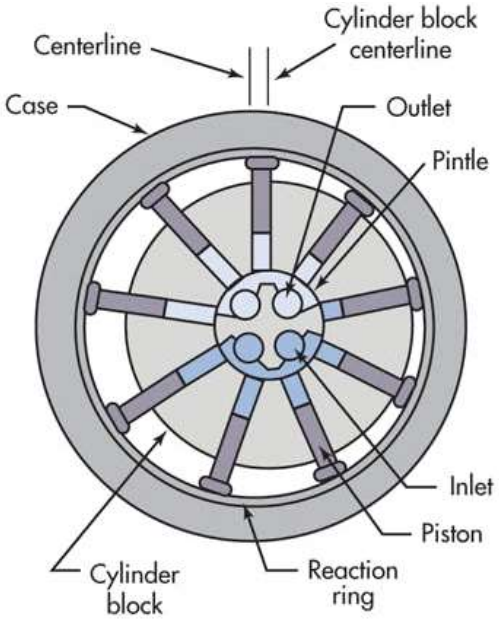
- Converts **mechanical energy → hydraulic energy.**
- Mechanical energy is usually driven by an **electric motor or engine.**
- Creates **flow** by drawing fluid from the reservoir and pushing it into the circuit.
- Pressure develops only when flow meets **resistance**

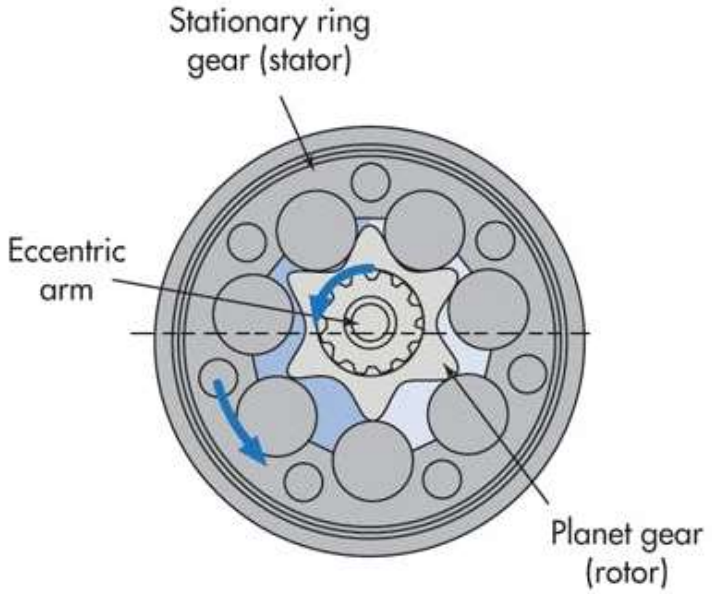


PC: https://www.powermotiontech.com/

Two type of pumps commonly used are:

- **Rotary Pumps**
  - Vane pumps
  - Gear pumps
- **Reciprocating Pumps**
  - Axial Piston Pump
  - Radial Piston Pump





PC: https://www.powermotiontech.com/

- Reciprocating pumps move liquids by changing the internal volume of the pump
- Require valves on the suction and discharge sides
- Pumping rates varied by changing the frequency or the stroke length

(Video)

- Rotary pumps transfer liquid through the action of a rotating mechanism (gear, lobe or vane) operating inside a rigid container
- Pumping rates varied by changing speed of rotor

video

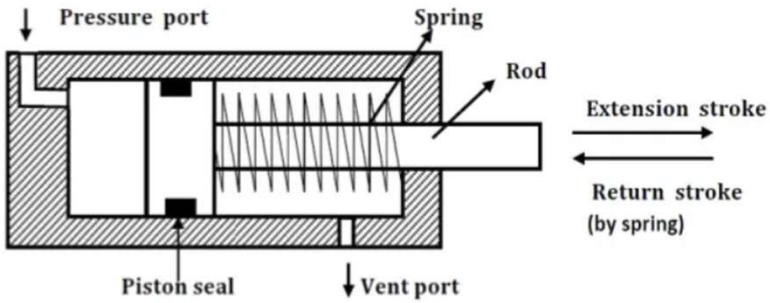
Hydraulic Valves

- Valves control **flow, direction, and pressure** of hydraulic fluid.
- Selected based on **type, size, actuation, and control method.**
- Major categories (based on primary function):
  - **Directional Control Valves (DCV)**
    - These valves control the direction of fluid flow or route the flow. (e.g. Check valves, solenoid valves ..)
  - **Pressure Control Valves**
    - These valves regulate fluid pressure
    - protect and regulate system pressure. (e.g Pressure Relief Valve)
  - **Flow Control Valves:**
    - These valves regulate the flow rate of hydraulic fluid
    - set actuator speed
    - E.g Needle valve, throttle check valve



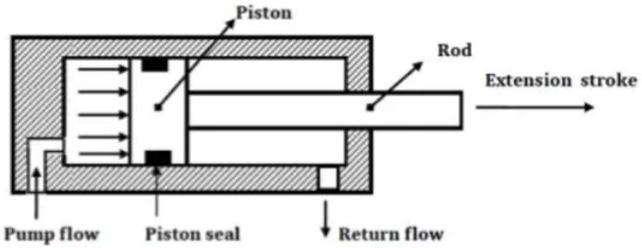
Hydraulic Cylinders (Linear Actuators)

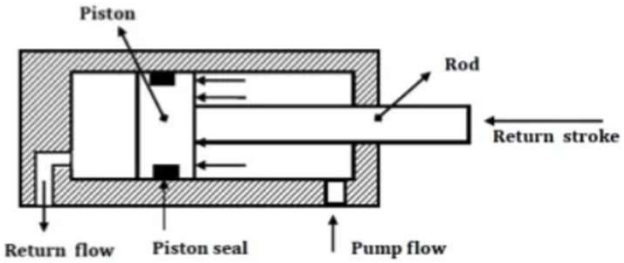
- Convert **fluid power → linear mechanical motion.**
- Provide **push or pull** along a straight line.



Types:

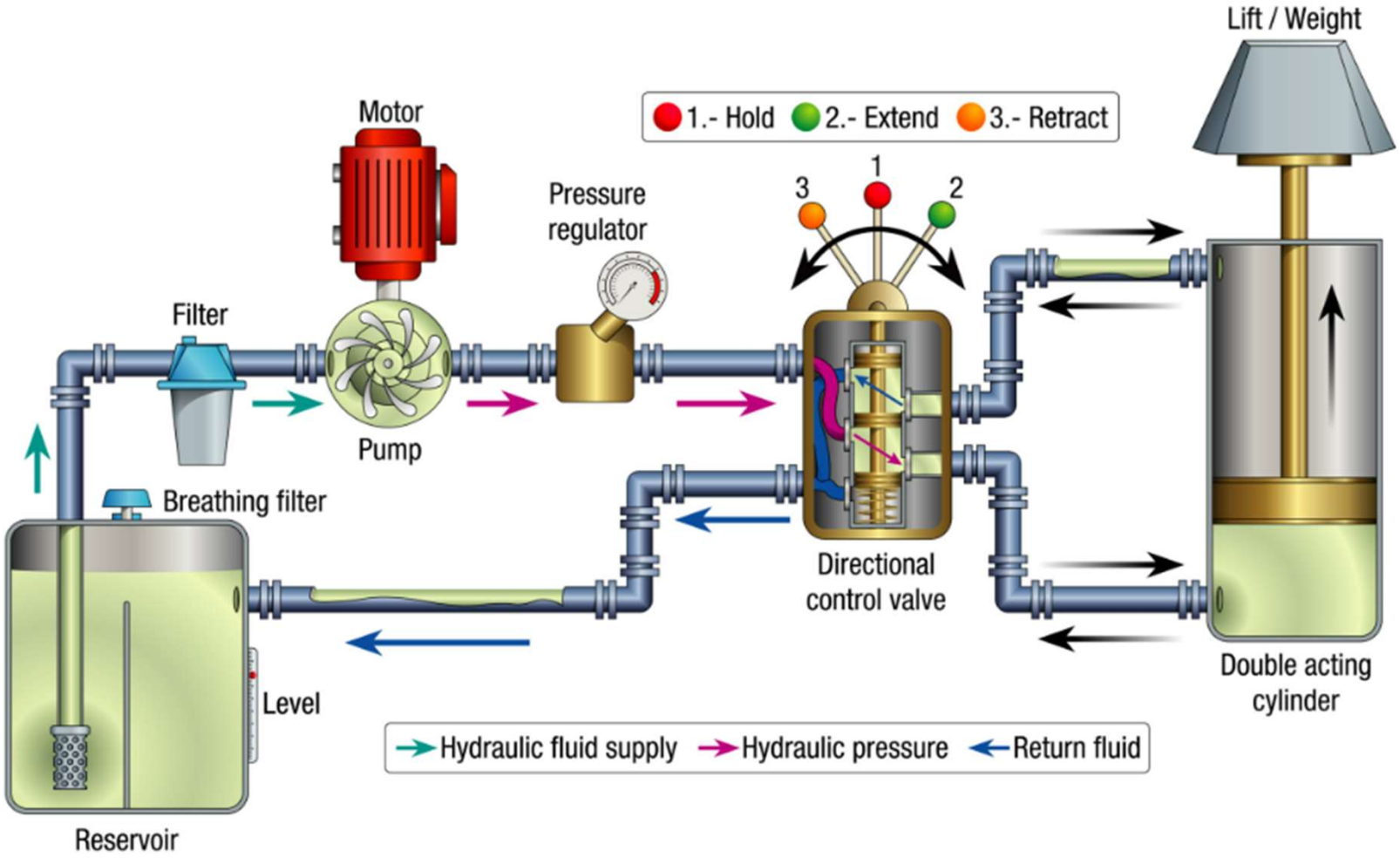
- **Single-acting cylinder:** Force in one direction (return by spring or gravity or other mechanism).
- **Double-acting cylinder:** Force in both directions.





Hydraulic Motors (Rotary Actuators)

- Convert **fluid power → rotary motion and torque.**
- A hydraulic rotor actuator (also known as a helical spline or rotary vane actuator) converts pressurized hydraulic fluid into precise rotational motion for high-torque applications
- **Types:**
  - Limited rotation (less than one turn).
  - Continuous rotation (full rotary motion).
- Used in: winches, conveyors, and wheel drives.


Lift / Weight
1.- Hold
2.- Extend
3.- Retract
Motor
Pressure regulator
1
2
3
Filter
Pump
Breathing filter
Directional control valve
Double acting cylinder
Level
Reservoir
Hydraulic fluid supply
Hydraulic pressure
Return fluid

Recap Quiz

Q. According to **Pascal's Law**, when pressure is applied to a confined fluid:
a) It is absorbed at the point of application
b) It is transmitted equally in all directions
c) It decreases with distance
d) It acts only in one direction

Q. In a hydraulic system, the **pump's primary function** is to:
a) Generate pressure
b) Create flow
c) Store energy
d) Control temperature